

Gauge and Homological Structures in Quantum Error Correction



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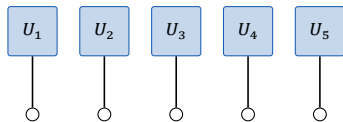
Based on arXiv:2511.15224, 2602.14499, and work in preparation

Outline

- **Part I:** Introduction
fault-tolerant logical gates; CSS codes as chain complexes
- **Part II:** Transversal Z -rotations and obstruction theory
arXiv:2602.14499 (PTEP 2026)
- **Part III:** Homological classification of logical diagonal gates
work in preparation with K. Ehara (UTokyo)
- Summary and outlook

Motivation: one mathematical framework

- This talk: **fault-tolerant logical gates**
- transversal or constant depth
(bounded error spread)
- without ancilla
(no space overhead)



transversal: no entangling gates

Goal: understand fault-tolerant logical gates
in a **unified mathematical framework**.

Fault-tolerant logical gates: what is known

- (Many) Examples: *CNOT* on CSS codes, *H* on self-dual codes; Steane code, Reed-Muller code, and many more
Shor 1996; Steane 1996; Knill, Laflamme, Zurek 1996; etc.
- No-go: no universal transversal set; locality-preserving $\Rightarrow$ level $\leq D$
Eastin, Knill 2009; Bravyi, König 2013; Pastawski, Yoshida 2015; Jochym-O'Connor, Kubica, Yoder 2018
- Algorithm: find the logical diagonal gates of a given code
Rengaswamy et al. 2020; Webster, Quintavalle, Bartlett 2023

So far: either a no-go theorem, or an answer code by code.
Is there a unifying viewpoint?

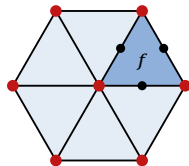
CSS codes: chain complex and homology

$$\begin{array}{ccccc} C_2 & \xrightarrow{\partial_2 = H_Z^T} & C_1 & \xrightarrow{\partial_1 = H_X} & C_0 \\ \text{Z check} & & \text{qubit} & & \text{X check} \end{array}$$

- Logical $Z, X \leftrightarrow$ first (co)homology

$$H_1 = \ker H_X / \operatorname{im} H_Z^T, \quad H^1 = \ker H_Z / \operatorname{im} H_X^T$$

- How about gates **beyond Pauli**?



face = C_2 (Z check), edge/qubit = C_1 ,
vertex = C_0 (X check)

Homological approach to logical gates

- **Cohomological invariants** (cup products, Steenrod squares, etc.) $\Rightarrow$ diagonal gates
Vasmer, Browne 2019; Barkeshli et al. 2024; Breuckmann et al. 2024;
Hsin, Kobayashi, Zhu 2024; Lin 2024; Golowich, Lin 2024
- **Chain maps** as logical CZ-type gates, organized into a hierarchy
Sahay, Long, Khemani 2026
- Many constructions, **no classification**

This talk: classify the logical diagonal gates on every CSS code

Part II: **transversal phase gates**

Part III: **all diagonal circuits**

(Clifford case, level 2, for simplicity; easily generalized to higher levels)

Part II

Transversal Z -rotations and obstruction theory

arXiv:2602.14499 (PTEP 2026, 083A03)

Transversal Z -rotations at level m

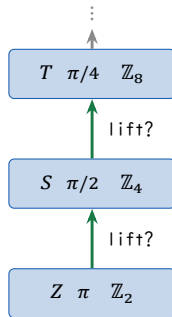
- Transversal Z -rotations at level m of the hierarchy

$$U(\theta) = \prod_{j=1}^n \left(Z_j^{1/2^{m-1}} \right)^{\theta_j}, \quad \theta \in \mathbb{Z}_{2^m}^n$$

- Level m : $\mathbb{Z}_{2^m}$ coefficients

$$\mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \cdots \rightarrow \mathbb{Z}_{2^m}$$

- Design test: hand me a code; is logical S transversal? ($m=2$ below)



Transversal phase gates: what is known

- **Weight-based criteria**: divisibility, tri-orthogonality
Ward; Bravyi, Haah; Haah; Rengaswamy et al.; Hu, Liang, Calderbank; Shi et al.; Jain, Albert
- **Parity constraints** across the Clifford hierarchy
Campbell, Anwar, Browne; Anderson et al.; Jochym-O'Connor, Kubica, Yoder; Camps-Moreno et al.
- **Structural generalizations**: self-dual codes, XP stabilizer formalism
Tansuwanont et al.; Webster, Brown, Bartlett

Asked in this part

A **closed form** for any CSS code: which gates, at which level?

Logicality for $\bar{S}$: two conditions on θ

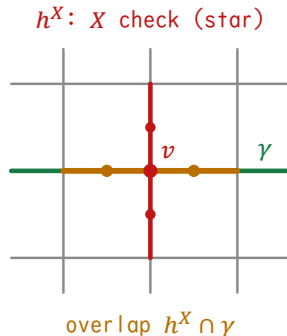
- Commutation with $S^X(h^X)$, h^X a row of H_X :

$$[U(\theta), S^X(h^X)]_g = \underbrace{e^{2\pi i \langle \theta, h^X \rangle / 4}}_{\text{phase}} \underbrace{U(2 \theta \circ h^X)}_{\text{Z-string}}$$

- Two conditions, **two moduli**:

- phase**: $H_X \theta = 0 \pmod{4}$
- Z-string = stabilizer**: $(H_X \odot K) \theta = 0 \pmod{2}$

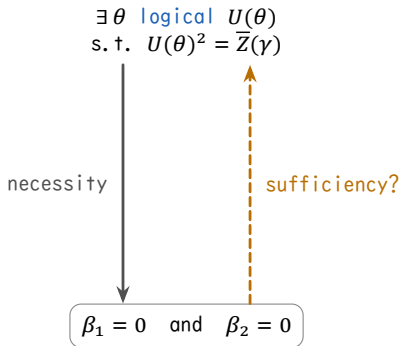
- Which logical Z ? $\Rightarrow$ one more constraint on θ, γ



$[A, B]_g$: group commutator; K = basis of $\ker H_Z$ stacked as rows; $\odot, \circ$: entrywise products

From θ to γ : two obstruction maps

- Which logical Z ? $U(\theta)^2 = \prod_i Z_i^{\theta_i} \stackrel{!}{=} \bar{Z}(\gamma)$
 $\theta = \gamma \pmod{2} \Leftrightarrow \theta = \gamma + 2\omega \pmod{4}$
- Substituting, **two conditions on γ**
 1. $(H_X \odot K)\gamma = 0 \pmod{2}$
 2. $\exists \omega, \frac{1}{2}H_X\gamma = H_X\omega \pmod{2}$
- Necessary; sufficient? $\Rightarrow$ **yes** (next)



Main theorem

Theorem (arXiv:2602.14499, level 2). For a CSS code, a transversal $\bar{S}$ on $\bar{Z}(\gamma)$ exists $\Leftrightarrow$ two obstruction maps vanish ($\beta_i = [0]$):

$$\begin{aligned}\beta_1([\gamma]) &:= [(H_X \odot K)\gamma] \\ \beta_2([\gamma]) &:= [H_X \gamma / 2] \quad \in \text{coker } H_X\end{aligned}$$

- Both maps (can be modified to) depend only on the homology class $[\gamma]$
 $\Rightarrow$ **homological criteria!**

β_2 is a Bockstein homomorphism

- $\bar{Z} \rightarrow \bar{S}$: lift coefficients $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$; the obstruction is the **connecting map**

$$\begin{array}{ccccccc} & & [H_X \gamma / 2] & & [\theta] & & [\gamma] \\ 0 & \longrightarrow & \mathbb{Z}_2 & \xrightarrow{\times 2} & \mathbb{Z}_4 & \xrightarrow{\text{mod } 2} & \mathbb{Z}_2 \longrightarrow 0 \end{array}$$


β_2 : Bockstein connecting map

lift exists $\Leftrightarrow \beta_2([\gamma]) = 0$ (keeping $[\gamma] = [\theta \bmod 2]$)

Transversal implementability = a **lifting problem in obstruction theory**.

Part II: Summary and Remark

Summary

- Implementability of a transversal **phase gate**
 $\Leftrightarrow$ **homological obstruction** to lifting coefficients

Remark

- Higher levels**: just a sequence of lifts $\mathbb{Z}_2^v \rightarrow \mathbb{Z}_2^{v+1}$ along $Z \rightarrow S \rightarrow T \rightarrow \dots$
- Known criteria**: divisibility, tri-orthogonality = necessary at $\theta = 1_n$ only

Next Beyond transversal:

homological classification of (FT) logical gates by S and CZ gates

Part III

Homological classification of logical diagonal gates

work in preparation with K. Ehara (UTokyo)

Logical gates by physical diagonal Cliffords

- Consider diagonal Clifford gates

$$U_b = \prod_i S_i^{b_{ii}} \prod_{i < j} CZ_{ij}^{b_{ij}}, \quad b = b^T, \quad b_{ii} \in \mathbb{Z}_4, \quad b_{ij} \in \mathbb{Z}_2$$

- Action on the computational basis: a $\mathbb{Z}_4$ -valued quadratic form

$$U_b |\gamma\rangle = i^{q_b(\gamma)} |\gamma\rangle, \quad q_b(\gamma) = \gamma^T b \gamma \bmod 4$$

- Questions:

1. When is U_b logical?
2. When is U_b a logical identity?

Logicality for U_b : two conditions on b

- Commutation with $S^X(h^X)$, h^X a row of H_X :

$$[U_b, S^X(h^X)]_g = \underbrace{i^{(h^X)^T b h^X}}_{\text{phase}} \cdot \underbrace{Z(b h^X)}_{\text{Z-string}}$$

- Two conditions, **two moduli** again:

(L1) **phase**: $(h^X)^T b h^X = 0 \pmod{4}$

(L2) **Z-string = stabilizer**: $b(\text{im } H_X^T) \subseteq \text{im } H_Z^T \pmod{2}$

Classification: quadratic forms on H^1

Theorem. On every CSS code,

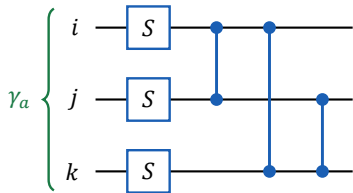
$$\{\text{logical } U_b\} / \{\text{logical id.}\} \cong \{\mathbb{Z}_4\text{-valued quadratic forms on } H^1\}$$

- **Realization:** with $\gamma_a \in \ker H_X$ a basis of H_1 , $\bar{Z}_a = Z(\gamma_a)$,

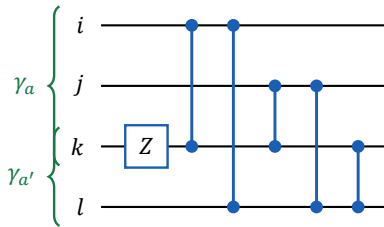
$$\bar{S}_a : b = \gamma_a \gamma_a^T, \quad \overline{CZ}_{aa'} : b = \gamma_a \gamma_{a'}^T + \gamma_{a'} \gamma_a^T$$

- Any other b with the same logical action differs by a **logical identity**

The circuits: $\bar{S}_a$ and $\overline{CZ}_{aa'}$



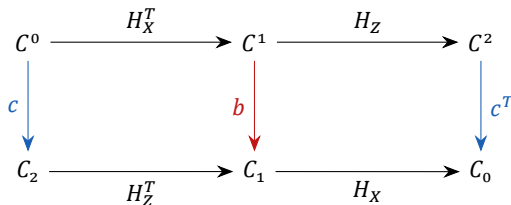
$\bar{S}_a$: S on every qubit of γ_a ,
 CZ on every pair



$\overline{CZ}_{aa'}$: CZ between γ_a and $\gamma_{a'}$,
 Z on the overlap

Chain map interpretation

- Logical b forms a **chain map**: $\exists c \in \mathbb{Z}_2^{m_Z \times m_X} \quad b H_X^T = H_Z^T c \pmod{2}$
cf. Sahay, Long, Khemani 2026



- Intuition:** $[U_b, X(\tilde{\gamma})]_g = i \tilde{\gamma}^T b \tilde{\gamma} Z(b \tilde{\gamma})$
 U_b logical $\Leftrightarrow [b \tilde{\gamma}] \in H_1$ for $[\tilde{\gamma}] \in H^1$
 $\Rightarrow b$ induces a linear map $H^1 \rightarrow H_1 =$ a **quadratic form on H^1**

The logical identities, completely

- Whole circuit set for a given **logical diagonal action**:

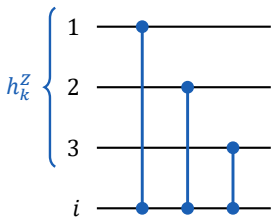
$$\underbrace{\{\text{logical } U_b\}/\{\text{logical id.}\}}_{= \text{quadratic forms on } H^1} + \underbrace{\{\text{logical id.}\}}_{\text{unknown}}$$

- Logical identity $\Leftrightarrow$ (L1') **phase**: $k_Z^T b k_Z = 0 \pmod{4} \quad \forall k_Z \in \ker H_Z$ and (L2)

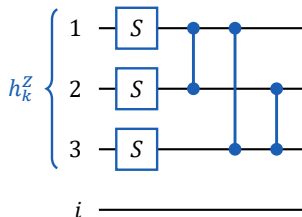
Theorem. On every CSS code, with h_k^Z the rows of H_Z ,

$$\{\text{logical id.}\} = \text{the group generated by } e_i(h_k^Z)^T + h_k^Z e_i^T, \quad h_k^Z (h_k^Z)^T, \quad \text{diag}(2h_k^Z)$$

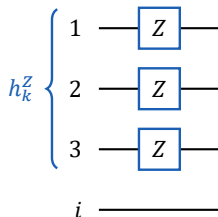
The circuits: logical identities



$e_i(h_k^Z)^T + h_k^Z e_i^T$:
controlled-Z stabilizer



$h_k^Z(h_k^Z)^T$:
 S on each qubit,
 CZ on each pair



$\text{diag}(2h_k^Z)$:
 Z stabilizer

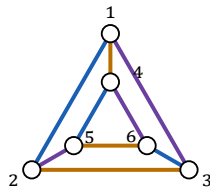
Depth of logical diagonal gates

- The generators are a **complete** set
⇒ all circuits for a given logical action!
- Which of them are **fault-tolerant**?
⇒ how much **depth** do we need?

Theorem. $\deg b :=$ degree of G_b , adjacency matrix b
(vertices = qubits, edges = CZ gates)

$$\text{depth of } U_b = \deg b + \text{const}$$

- **Proof:** depth = # of colors for **edge coloring**



$$b = \begin{pmatrix} * & 1 & 1 & 1 & 0 & 0 \\ 1 & * & 1 & 0 & 1 & 0 \\ 1 & 1 & * & 0 & 0 & 1 \\ 1 & 0 & 0 & * & 1 & 1 \\ 0 & 1 & 0 & 1 & * & 1 \\ 0 & 0 & 1 & 1 & 1 & * \end{pmatrix}$$

one color = one layer of CZ

Part III: Summary and Remark

Summary

- On every CSS code, $\{\text{logical } U_b\}/\{\text{logical id.}\} \cong$ quadratic forms on H^1
- Logical identities generated by the Z checks $\Rightarrow$ **all circuits** for a given logical action
- Depth of $U_b = \deg b + \text{const}$

Remark

- **Higher levels**: m -linear forms on H^1
- **Fault tolerance**: error spread of bounded-degree circuits
- **Gauge-field origin** of these gates (Haruna 2025, backup)

Summary and outlook

Summary

- **Part II:** implementability of a transversal phase gate
 $\Leftrightarrow$ homological obstruction to lifting coefficients
- **Part III:** logical diagonal gates
 = quadratic forms on H^1 + logical identities generated by Z checks
- **Gauge fields:** these gates have a gauge-theoretic origin
 on the chain complex

Towards a theory of FT logical operations

- **Fault tolerance** in the language of homology and beyond:
which representatives are cheap, and why
(cf. the talks by Jonathan, ...)
- **Beyond qubit CSS codes** (non-CSS, qudit, non-Abelian codes):
representation theory in place of Abelian cohomology
(cf. the talks by Anasuya, ...)
- **Fault-tolerant protocols with ancillas**: CPTP maps and homology?
generalized lattice surgery $\leftrightarrow$ mapping cone
(cf. the talks by Sunny, Dominic, ...)

Homology and beyond: a **unified framework** for FT logical operations

Thank you

Backup

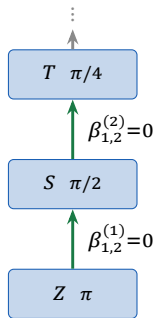
Climbing the Clifford hierarchy

- Same argument climbs the hierarchy $Z \rightarrow S \rightarrow T \rightarrow \dots$: one lift step $\mathbb{Z}_{2^v} \rightarrow \mathbb{Z}_{2^{v+1}}$ per level
- At level m the gates form

$$V_{\text{LD}}^{(m)} \cong H^1 \otimes_H S_m, \quad S_m = \bigoplus_{v=1}^m 2^{v-1} (\ker H_Z)^{\otimes_H (v-1)}$$

$\otimes_H$: Hadamard (entrywise) product of modules

- Divisibility, tri-orthogonality =
necessary conditions at uniform angle
part of the logicity condition for $\theta = 1_n$;
not sufficient in general



Touchstones

Code / criterion	In the classification
Steane $[[7,1,3]]$	transversal S : $\beta_1^{(1)} = \beta_2^{(1)} = 0$ yes ; transversal T : $(\beta_1^{(2)}, \beta_2^{(2)}) \neq 0$ no, even for non-uniform θ
$[[15,1,3]]$ Reed-Muller	transversal T : $\beta_{1,2}^{(1)} = \beta_{1,2}^{(2)} = 0$ yes

Non-existence of a gate is now **certified by computing an obstruction**, for one code or for a whole family.

Steane $[[7, 1, 3]]$: transversal S yes, T no

- Self-dual CSS ($H_X = H_Z = H$, the $[7, 4, 3]$ Hamming code):
a single logical class

gate	lift	obstruction	transversal?
$S(\pi/2)$	$1 \rightarrow 2$	$\beta_1^{(1)} = \beta_2^{(1)} = 0$	yes
$T(\pi/4)$	$2 \rightarrow 3$	$(\beta_1^{(2)}, \beta_2^{(2)}) \neq 0$	no

- S : lift $\hat{\theta} = 3 \cdot 1_7$ gives $\langle \hat{\theta}, 1_7 \rangle = 1 \pmod{4}$, so $\prod_i S_i^\dagger$ is logical S
- T : every logical $\theta^{(3)} = 2c_0 1_7 + 4 \sum_i c_i h_i$ is even,
so $\langle \theta^{(3)}, 1_7 \rangle \neq 1 \pmod{8}$: no transversal T , even for non-uniform $\theta \in \mathbb{Z}_8^7$

[[15, 1, 3]] Reed-Muller: transversal T

- Punctured Reed-Muller code; physical T on every qubit $\rightarrow$ logical $T^\dagger$
- In this framework: uniform angle $\theta_i \equiv 1$, level $m = 3$, $\mathbb{Z}_8$ coefficients
- The gate exists, so by the theorem the ladder vanishes: $\beta_{1,2}^{(1)} = \beta_{1,2}^{(2)} = 0$
- **Tri-orthogonality** (even overlaps of rows of H_X , pairwise and triple)
= part of the logicality condition for $\theta = 1_n$ at level 3

Tri-orthogonality is a necessary condition at uniform angle: it arises from logicality, and does not characterize the vanishing of the obstruction maps.

Known criteria as necessary conditions

At uniform angle $\theta = 1_n$, the known criteria are **part of the logicality condition**:

Known criterion	In this framework
Divisibility : every X stabilizer has weight divisible by 2^m	first block of the logicality condition, $H_X 1_n \equiv 0 \pmod{2^m}$: necessary at level m , not sufficient
Tri-orthogonality : overlaps of rows of H_X even, pairwise and triple	triple Hadamard-product block at level 3, $(H_X \odot H_X \odot H_X) 1_n \equiv 0 \pmod{2}$: necessary, not sufficient
General θ	further constraints from the higher Hadamard-product blocks; the lift to a finer angle is decided by β_1, β_2

Divisibility and **tri-orthogonality** are necessary conditions that emerge from logicality at $\theta = 1_n$; they do **not characterize the obstruction maps**.

Relation to Webster-Quintavalle-Bartlett

Their XP-stabilizer algorithm (arXiv:2303.15615) classifies transversal diagonal logicals of any stabilizer code.

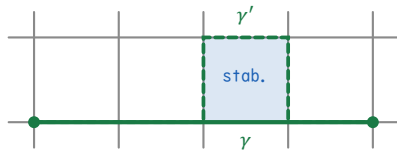
At each level m , the two classifications coincide: $V_{\text{LD}}^{(m)} = V_{\text{L}}^{(m)} / V_{\text{LI}}^{(m)}$.

	WQB (2303.15615)	This work
Method	implicit kernels, per-code algorithm (Howell / Smith normal form over $\mathbb{Z}_2^m$)	a closed form for an arbitrary CSS code
Output	a generating set, code by code	obstruction maps β_1, β_2 on homology
Scope	multi-qubit controlled-phase; non-CSS via embedding	inter-level lifting; natural under maps of codes (families, covers)

Same classification, complementary content: WQB compute it code by code; the homological form exposes the structure and carries the lifting obstruction.

β_1, β_2 live on homology

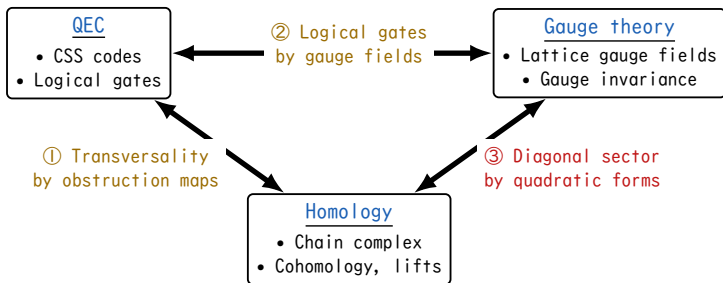
- β is computed from one representative γ
- $\bar{Z}$: $\gamma \bmod \text{stabilizers}$
(many γ , one logical)
- Lemma: a stabilizer shifts γ , not β
 $\Rightarrow \beta_1, \beta_2$ live on $\bar{Z}$, not on γ



$$\gamma' = \gamma + \partial(\text{stab.})$$

$$[\gamma'] = [\gamma] \Rightarrow \beta(\gamma') = \beta(\gamma)$$

Three structures, one chain complex



- ① Transversal Z-rotations: which logical gates, at which level (2602.14499) Part II
- ② Gauge fields on the chain complex: closed-form logical gates (2511.15224) backup
- ③ Diagonal sector at level 2, and its constant-depth part (in preparation) Part III

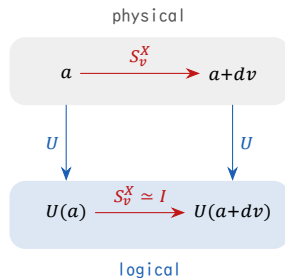
Gauge fields on the chain complex

- Qubit $e_j \mapsto$ projector-valued cochains
(electric, magnetic)

$$a(e_j) = \frac{1}{2}(I - Z_j), \quad b(e_j) = \frac{1}{2}(I - X_j)$$

- X -check conjugation = **gauge transformation**
 $a \rightarrow a + dv$
- Logical operator = **gauge-invariant** function of a, b
replacement principle: write the gate in Z ,
then $Z \rightarrow \bar{Z}(\gamma)$

$$\bar{Z}(\gamma) = e^{i\pi a(\gamma)}, \quad \bar{S}(\gamma) = e^{i\pi a(\gamma)^2/2}, \quad \overline{CZ}(\gamma_1, \gamma_2) = e^{i\pi a(\gamma_1)a(\gamma_2)}$$



Closed form for the full Clifford group and T on **any CSS code** (next slide); the action depends only on the (co)homology class.

Full Clifford + T on general CSS codes

- Same recipe, full Clifford + non-Clifford T ; $\tilde{\gamma}$ the dual (X -type) loop

Logical gate	Gauge field expression
$\bar{S}(\gamma)$	$\exp(i\pi a(\gamma)^2/2)$
$\bar{H}(\gamma)$	$\bar{S}(\gamma) \exp(i\pi b(\tilde{\gamma})^2/2) \bar{S}(\gamma)$
$\bar{CZ}(\gamma_1, \gamma_2)$	$\exp(i\pi a(\gamma_1)a(\gamma_2))$
$\bar{T}(\gamma)$	$\exp(i\pi (2a(\gamma)^3 - 3a(\gamma)^2 + 2a(\gamma))/4)$

- No special structure assumed: applicable to general CSS codes

From one gate to the whole sector

- Every diagonal gate is one master formula, $P_j^- = a(e_j)$:

$$U_{\text{diag}} = \exp\left(i f(P_1^-, \dots, P_n^-)\right), \quad f \text{ a polynomial}$$

- Level = angle and degree of f together:
 Z, S, T linear at $\pi, \pi/2, \pi/4$; CZ
quadratic at π ; CCZ cubic
- Part II: the transversal slice, f
linear at every angle, classified
completely
- Gauge fields: one representative per
logical gate, e.g.

$$\overline{CZ}(\gamma_1, \gamma_2) = \exp(i\pi a(\gamma_1) a(\gamma_2))$$

a rank-2 quadratic form in a ,
of weight $|\gamma_1||\gamma_2|$

Problem (B), made precise

Fix the level: f quadratic, i.e.
all circuits of S and CZ .

Which of them are logical, which
logical gates do they give, which
are identities, and which have a
constant-depth representative?

The answer: quadratic forms on H^1 , and sparse
symmetric chain maps. Part III.

Lifting a symmetric chain map to a logical b

- **Lemma.** $\beta: C^\bullet \rightarrow C_{2-}^\bullet$ a symmetric chain map (mod 2). Then there is a logical b with $b \equiv \beta \pmod{2}$ (diagonal included), $\deg b \leq \deg \beta + 1$, and the same induced pairing $\bar{B}$ on H^1 .
- **Proof.** Lift the diagonal of β to $\{0,1\}$: b_0 . (L2) is mod 2, so b_0 satisfies it.
- On $H = \text{im } H_X^T$: $h^T \beta h' = 0$ for $h, h' \in H$, so by polarization q_{b_0} ($q_b(h) := h^T b h$) is **additive on H** , and $q_{b_0}(h) \equiv h^T \beta h \equiv 0 \pmod{2}$: values are even.
- So $f(h) := q_{b_0}(h)/2 \pmod{2}$ is $\mathbb{Z}_2$ -linear on H ; extend to $\mathbb{Z}_2^n$ as $f(h) = \ell^T h$.
- Set $b := b_0 + \text{diag}(2\ell)$. Then $q_b(h) = q_{b_0}(h) + 2\ell^T h \equiv 2f(h) + 2f(h) \equiv 0 \pmod{4}$: (L1) holds, (L2) unchanged, and the diagonal shift raises row weights by at most 1.

(L1) never obstructs a sparse chain map: the mod-2 object decides the depth, the $\mathbb{Z}_4$ diagonal is **fixed afterwards** at the cost of one layer of Z .

Why the pairing must include the diagonal

- Polarization on the diagonal: $x \oplus x = 0$ and $\bar{q}(0) = 0$ give $\text{pol}_{\bar{q}}(x, x) = -2\bar{q}(x)$, i.e. $\bar{B}(x, x) = \bar{q}(x) \bmod 2$
- In the normal form: $\bar{B}(v_c, v_c) = s_c \bmod 2$ and $\bar{B}(v_c, v_d) = t_{cd}$: the diagonal of $\bar{B}$ records the **parity of the $\bar{S}$ coefficients**
- Prescribing $\bar{B}$ with zero diagonal excludes odd s_c ; $\bar{B}$ cannot see $s_c = 0$ versus 2, which is only the logical Pauli $\bar{Z}$ freedom $\text{Hom}(H^1, 2\mathbb{Z}_4)$
- $[[4, 2, 2]]$ count: of the 512 logical b with $\bar{B}_{12} = 1$, 384 carry an $\bar{S}$ admixture; the 128 with zero diagonal are exactly the $\overline{CZ}_{12}$ coset

Off-diagonal data specifies the CZ sector; the
diagonal specifies whether an S sneaks in.
Both are mod-2 data of the chain map.

Relation to Sahay, Long and Khemani

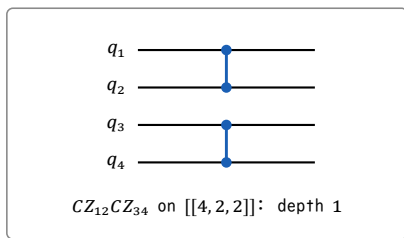
- arXiv:2609.02999: a [chain map hierarchy](#) for logical CZ-type gates
- Shared dictionary: chain map = CZ circuit; null-homotopy = trivial gate; sparse representative = constant depth; mod 2 decides the depth
- Overlap: their mod-2 boundary group = our trivial sector mod 2, $\langle g^1, g^2 \rangle$ (g^1, g^2, g^3 : the three generators of the logical identities, in the order of p21)

Added here

- exact logicality criterion (L1) + (L2), no lift of the chain complex needed
- the trivial sector completely, with its $\mathbb{Z}_4$ structure (g^2, g^3)
- the group isomorphism $V_{LD} \cong \text{QF}(H^1)$ and the gate dictionary
- when the diagonal can be corrected afterwards, **diagonal-inclusive pairing** plus the lifting lemma; depth bound $\deg \beta + 2$; level m by finite polarization

Examples, machine-checked

Target	Result
$[[4,2,2]], \overline{CZ}_{12}$	minimal degree 1: $CZ_{12}CZ_{34} = \overline{CZ}_{12}\overline{Z}_1$ (exhaustive, all 16384 classes)
$[[4,2,2]], \overline{S}_1$	minimal degree 2: $S_1S_3CZ_{13} = S(\hat{g}_1)$, a weight-2 divided square
two Steane blocks, $\overline{CZ}$	transversal CZ , $b = \text{antidiag}(I)$, degree 1: self-duality gives (L2), the CSS condition gives (L1)



$H_X = H_Z = (1111)$; $\hat{g}_1, \hat{g}_2$ dual to the two logical classes

On $[[4,2,2]]$, $\overline{S}$ is **one step deeper** than $\overline{CZ}$; $[[4,2,2]]$ is the minimal model of the gap between the CZ and S sectors.